A new 194-state PA-consistency machine

PA194QNS, a complete comparison of the current corpus, and structural reduction research

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Abstract

No verified machine below 194 working states was obtained. The requested same-size alternative is supplied by **PA194QNS**, a deterministic binary machine on one two-way tape, started on the all-zero tape, with 194 working controls and 388 transitions. It combines the Q axiom emitter, no-push modus ponens, and the symmetric implementation of the original sum-based Shift test. These mechanisms are attributed to their earlier packets; their exact composition, source equivalence, transition table, novelty comparison and fresh verification are the new work.

The source has 3,651 primitive instructions. Its first 3,649 instructions are literally identical to the pinned QN source, and the original five-instruction tail is replaced by two instructions. A simple invariant and symmetry of destructive equality give a PA-formalizable finite-trace equivalence. The physical count is $135 + 67 - 8 = 194$. All 4,751 arbitrary-register source outcomes pass both symbolic checks; the independent literal checker rederives its four list-sweep lemmas from the actual rows. Exact blank initialization takes 58,004 steps and is checked on the complete tape.

The resulting ordinary mathematical argument, composed with the attributed QN calibration, gives $\mathsf{PA} \vdash (\neg\,\mathrm{Halt}(\mathrm{PA194QNS}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathsf{PA}))$. This is not a newly compiled end-to-end Lean or Rocq theorem, and no completed iPRA consistency reduction is claimed. The comparison covers all 54 current deliveries and 39 existing 194-state graph classes, including auxiliary and rejected graphs. The archive supplies the complete machine, reproducible code, certificates, source proofs, search evidence and this report. A separate mathematical coding experiment is explicitly distinguished from the accepted 194-state result.

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1 Result and the meaning of the claim

The principal table is `construction/machine/machine.tm`. Its working states are $0, \ldots, 193$, its initial state is 0, and 194 is the separate terminal halt label. The head starts at coordinate zero on an all-zero two-way infinite binary tape. Every action writes one bit and moves exactly one square left or right. No stationary moves, nonblank input or change of counting convention is used.

Theorem 1.1 (The delivered same-size alternative)**.** *For the exact table in Appendix B,*

$$\boxed{\mathsf{PA} \vdash (\neg\, \mathrm{Halt}(\mathrm{PA194QNS}, \mathrm{blank}) \iff \mathrm{Con}(\mathsf{PA}))\,.}$$

Here Halt is the ordinary arithmetization of a finite literal-machine run, and $\mathrm{Con}(\mathsf{PA})$ abbreviates the absence of a finite PA proof of $0 = 1$ in a fixed standard proof calculus. The new part of the argument is an explicit finite-trace transport to the pinned QN implementation; its ordinary PA consistency calibration is an attributed dependency. Section 3 states the logical and proof-word interfaces rather than inferring them from test execution.

The biconditional is a statement about what PA proves. To infer actual nonhalting in the standard model one additionally assumes PA's consistency. The construction does not prove PA consistent, and it does not establish that 194 is minimal. The unsuccessful searches below have precisely stated finite or bounded scopes.

The new machine differs from every existing 194-state literal graph even after arbitrary state renaming, change of spanning root, left/right reflection and global symbol interchange. More substantively, it changes a recurring comparator computation relative to QN, and retains a different quantifier-distribution head from the old sum-NS implementation. It is a new combination of known mathematical and implementation mechanisms, not a claim to have invented those mechanisms individually.

2 What the complete comparison establishes

The comparison is frozen at ProveIt commit

`317b07268681b44ff352d173a162e2a8c241b2fe`.

It covers all 51 delivered packets filed under `Computability/BusyBeaver/Research/PA/docs`, the three current incoming archives, the primary PA documentation and imported PA196/PA209 sources. The ZFC and ConciseIPRA projects were examined for transferable mechanisms. The three additional archives are `PA194NS_research.zip`, `pa194haz_research_2026-11-14.zip`, and `pa194qn_research_2026-11-13.zip`. Thus there are **54 deliveries** in the narrative ledger. Immutable paths, packet summaries and provenance appear in `comparison/packet_ledger.csv` and its JSON counterpart.

2.1 Literal tables, graph classes and evidence status

The new scanner reads every tracked `.tm`, `.txt`, `.csv` and `.tsv` file in the two requested locations, and recursively reads ZIP archives. It visits identical ZIP payloads once. A recognized table is a complete numeric five-column binary transition table with left/right moves and one undefined target serving as halt. The resulting inventory is:

Item	Count
Delivery packets	54
Recognized literal-table occurrences	1,402
Distinct table byte strings	721
Complete action graphs with a spanning root	497
194-state action graphs	39
195-state action graphs	19
196-state action graphs	99

The scanner examines 3,480 candidate text occurrences and 17 distinct ZIP bodies reached through 29 references. The 194-state portion contains 179 occurrences and 94 byte presentations. Two disconnected synthetic fixtures with six and nine controls are recorded separately; another synthetic fixture has no terminal target and fails the recognizer. There is *no complete spanning-root graph with 7–193 working states* in the recognized corpus. This is an inventory result, not a minimality theorem.

The graph comparison is stronger than a comparison of filenames or hashes. For every possible root reaching the whole table it serializes the complete action graph in deterministic traversal order. It then compares the canonical strings, also allowing global tape reflection and interchange of the two symbols. All 39 current 194-state classes remain distinct under these transformations. Symbol interchange is included only as a stronger structural exclusion test: it does not by itself preserve a distinguished zero blank. The new machine is compared against *all* these classes, including auxiliary and rejected objects.

Of the 39 existing classes, **26 are presented as constructions**, nine are auxiliary or search objects, and four are explicitly rejected implementations. These labels report the supplied evidence; they do not assert that all historical proof obligations were independently discharged in this investigation. Appendix A lists each class. Every table, its identity, its member paths and its evidence classification can also be inspected in the machine-readable census.

Old reports have different occurrence totals because repository filing has removed repeated table copies, and older archives recursively quoted one another. Those totals must not be added. The current scanner deliberately does not infer tables hidden only in arbitrary program expressions, JSON, prose or uninstantiated search parameters. The packet-by-packet narrative review complements that explicit literal-table boundary.

2.2 The main 194-state mechanisms

Family	What changes, and what its comparison requires
PA194, AD, 68/8	The separated diagonal head and ten-entry AX interface are shared. The primary budget is $134 + 67 - 7$; AD changes register placement, and 68/8 trades an extra decoder reader for an extra compatible fold. Their resolved sources or source conjugacies must be compared, not only the counts.
C, CS	Successful MP leaves its result in the cache; Shift tests the cache alone. CS changes the destructive comparator. These have 3,649 and 3,646 source sites. Their proof-word interface differs from the original sum-guard controller.

Family	What changes, and what its comparison requires
N, H, HS	N omits a successful-MP zero push but retains the sum guard and visible result. H/HS combine no-push MP with a cache-only guard. N's successful result is $(b, w, b, 0)$; the retained offset and missing zero must be handled by the proof compiler.
M4R, M5, M6; N5, N6	A multiplier-decoder reassociation exposes another history fold. M5/M6 change formula coefficients to 5/6; the stack and scheduler still use Cantor pairing. Their hereditary formula-and-label recoding is not order preserving. N5/N6 add no-push MP.
J, D, Q, QN	J changes return organization; D emits a diagonal instance indexed by another register. Q exchanges the curried premises of a quantifier-distribution axiom. QN adds no-push MP to Q. Logical and command interfaces must both be translated.
CZ, CSZ/X, A, F, R, ZR and auxiliaries	Known-zero decrements, scanner choices and source reassociations change physical return paths. Their local invariants need a separate blank-start check. Some source-valid changes fail during initialization.
HA, HAZ	These combine the older MPH cached-result interface with physical A/D placement; HAZ also changes the final AX return. Their sum guard differs from the cache-only H/HS family despite overlapping names.
SUMS, incoming sum-NS	These retain the original sum guard but compare its two operands in reverse order, removing a transfer loop. A new one-bit history fold pays for the trailer reader. The incoming sum-NS additionally omits the MP push.

2.3 Duplicates and ambiguous names

Several apparent additions to the frontier are the same literal graph: five principal cached-controller deliveries repeat PA194C; PA194X equals PA194CSZ; the filed PA194HS equals the lowercase packet's PA194NS; and the two N6 deliveries contain the same PA194N6. The *uppercase* incoming `PA194NS_research.zip` uses PA194NS for a different machine, preserving the sum guard rather than testing the cache alone. Its dense digest begins `eef9329b`; the filed HS/NS digest begins `19935635`. HAZ's preliminary H is the old MPH auxiliary, not the H from the HS packet. Full paths and exact bytes resolve these collisions; names alone do not.

2.4 The larger machines

The 19 graphs at 195 states include the original bundled-diagonal PA195, the separated-head 195S, the cached 195C, the scanner variant M195S8, diagonal-combined PA195DC, quotient parents and rejected or auxiliary experiments. Two latest QN-tail experiments are physically checked but are not supplied with complete new consistency calibrations. Thus 19 is not a count of 19 independently established consistency theorems.

The 99 graphs at 196 states include the imported PA196 and later logical, register, controller and decoder variations. A shorter direct-negation source has a verified 197-state allocation: source length is not the physical objective. The 197–207-state graphs mostly record larger experiments or ancestors. There are 142 classes at 208 states, predominantly U208 research variants and quoted inputs; the principal U208 reductions use entry, finite-history and contextual folds of an older U209 implementation.

Imported PA209 and U209 are distinct source lineages. The census retains all recognized ancestors through 222 states and their exact paths.

2.5 A useful negative control

Three distinct comparator-return mutations independently halt after exactly **1,361 blank-tape steps**. The fresh simulator and their identities are included in `comparison/verify_negative_examples.py` and `comparison/early_halt_rechecks.json`. Their source-level zero invariants do not rescue a changed initialization. This is why a successful source contract and a full blank initialization are both required below. The fourth rejected 194-state graph is the shortened seven-zero scanner, whose supplied source-address contract fails; no new complete replay of that older failure is claimed here.

2.6 The formal proof boundary

The current PA project explicitly leaves its machine-specific iPRA consistency reductions open, while providing substantial Lean source, quotient, zipper, scheduler and clock infrastructure [1]. The delivered papers separately give ordinary mathematical arguments intended to be formalizable in PA. These are different claims. Absence of a newly compiled Lean endpoint does not disprove an ordinary paper proof; standard-natural-number simulation and a conditional arithmetic endpoint also do not discharge the missing iPRA reductions. This report proves its new transport and identifies the inherited PA-calibration arguments rather than silently upgrading one proof level to another.

3 Logical background and the inherited PA calibration

The numerical syntax uses Cantor pairing $p(a,b) = (a+b)(a+b+1)/2 + b$ and constructors

$$I(A,B) = 4p(A,B), \quad M(a,b) = 4p(a,b)+1, \quad U(a,B) = 4p(B,a)+2, \quad E(a,b) = 4p(a,b)+3.$$

Code zero denotes truth, code one denotes the chosen contradiction, and $N(B) = I(B,1)$. Write $A \to B$ for $I(A,B)$, and use the usual classical abbreviations for conjunction and existential quantification. The six universally instantiated heads, in their order of emission, are

$$\begin{aligned}
h_1(D,A,H) &= (\exists x_D\, U(D,H)) \to H,\\
h_2(D,A,H) &= U(D,U(A,H)) \to U(A,U(D,H)),\\
h_3(D,A,H) &= M(H,D) \to (E(H,A) \to M(A,D)),\\
h_4(D,A,H) &= E(H,D) \to (M(A,H) \to M(A,D)),\\
h_5(D,A,H) &= \exists x_D\, E(H,D),\\
h_6(D,A,H) &= U(D,A) \to \big(U(D,A\to H) \to U(D,H)\big).
\end{aligned}$$

The Q emitter replaces just the last of these by

$$h_6^{\mathrm{Q}}(D,A,H) = U(D,A\to H) \to \big(U(D,A) \to U(D,H)\big).$$

All other axiom heads, the diagonal schema $\Gamma(B) = B \to U(B,B)$, the guarded separation gate and the inference rules are retained. The propositional head is

$$\rho(D,A,H,d) = ((H\to d)\to D)\to((D\to H)\to(A\to H)), \qquad d \le D.$$

The guarded gate is $G(a,b,c) = N(V(a,b,c) \to Q(a,b,c))$ for $c \neq a, b$, with a failed guard producing zero, where

$$\begin{aligned} V(a,b,c) &= M(a,b) \to U(c, M(a,b)), \qquad P(a,b,c) = c \to M(c,b), \\ S(a,b,c) &= U\big(b, (M(a,b) \to P(a,b,c)) \wedge (M(c,b) \to c)\big), \\ Q(a,b,c) &= U(c, S(a,b,c) \to N(P(a,b,c))). \end{aligned}$$

The rules are modus ponens and generalization of a theorem $B \mapsto U(i,B)$, together with truth. Proofs are finite trees with explicit axiom parameters and inference annotations.

3.1 Why the exchanged head preserves the calculus

The unchanged dominated implication basis derives the K and S schemata, hence the uniform exchange theorem

$$(P \to (Q \to R)) \to (Q \to (P \to R)).$$

The relevant supplied Lean source is `M6/Logic/RestrictedImplication.lean`; its hypotheses are modus ponens, the dominated row, reflexivity of the numerical bound and propagation of bounds into either implication argument [5]. No version of the sixth head is used to obtain exchange. Instantiating exchange once derives either sixth head from the other. Recursively replacing the changed axiom leaves therefore gives primitive-recursive proof translations in both directions, preserving conclusions literally. PA verifies these translations by induction on finite derivations.

The new `proofs/exchange_certificate.json` is an additional closed 61-node K, S, MP derivation: 17 K instances, 14 S instances and 30 modus-ponens steps. Its small replay checker validates the published DAG separately from the bracket-abstraction program that generated it. This certifies the finite exchange template; the derivation of K, S from the dominated numerical basis remains the identified inherited theorem.

3.2 The no-push command interface

Let $[a;w] = p(a,w)$, with total Cantor projections. In particular $[0;0] = 0$ and both projections of zero are zero. A command store is (t,w,d,h): current formula, numerical stack, offset and cached formula. For $y = d + x$, QN and QNS implement

$$\mathsf{MP}_x(t,w,d,h) = \begin{cases} (y,w,y,0), & I(t,y) = h, \\ (0,w,y,0), & I(t,y) \neq h, \end{cases}$$

and

$$\mathsf{S}(t,w,d,h) = \begin{cases} \mathsf{halt}, & t + h = 1, \\ (h, \mathrm{tl}(w), d, \mathrm{hd}(w)), & t + h \neq 1. \end{cases}$$

The altered physical comparator preserves this entire command, including its original sum guard. It does not replace it by a cache-only test.

With $D = d + x$ and $H = h + y$, the axiom command has the same ten-entry interface:

$$\mathsf{AX}_{A,x,y}(t,w,d,h) = \big(g(H,D,A), [q_1, \ldots, q_9, t; w], 0, 0\big),$$

where $q_1 = \Gamma(A)$, $q_2 = h_6^{\mathrm{Q}}$, the next five entries are $h_5, \ldots, h_1$, $q_8 = \rho(D,A,H,d)$ and $q_9 = U(d,h)$. Each freshly generated code $g, q_1, \ldots, q_9$ is different from one; the saved incoming top t is unrestricted.

The complete proof compiler must reserve the zero that successful MP no longer pushes. The exact all-store Push, Pop, Discard, selector, generalization and MP contracts are retained in `construction/references/logical/pa194n_command_proof.tex`. They use total projections rather than treating the numerical stack as a free list, and provide a halting proof word from every incoming store for each derivation of code one. Cofinal axiom padding makes the unchanged Cantor scheduler eventually execute such a word.

3.3 The two inherited syntactic reductions

The separate-head PA calibration has two distinct directions [2, 3]. For completeness, the diagonal clause recovers the reverse membership rewriting and neighborhood-equality interfaces used by the original PA-refutation compiler. This produces a proof of code one in the numerical calculus from each PA refutation. For soundness, translate membership as finite binary membership, equality as equality and U as quantification. PA supplies finite binary comprehension by induction, via a primitive-recursive *proof-producing* template. Code freshness justifies the diagonal axiom. The $a = b$ separation case and the zero-code alias $I(0, 0) = 0$ are handled separately in those supplied arguments.

Annotate each live theorem code and canonical stack entry with a PA proof of its translation. The command operations preserve this annotation, and a halt gives a PA refutation. This uses finite syntax and proof checking; it does not introduce an arithmetic truth predicate or appeal to PA reflection. Together these syntactic reductions give the inherited PA-provable equivalence for QN. The new physical and source transport is given in full next.

4 PA194QNS: the certified fallback construction

The machine PA194QNS composes the supplied Q axiom emitter, the supplied no-push modus-ponens interface, and the supplied SUMS implementation of the original sum-based Shift guard. None of those components is claimed to be new individually. The new object is their exact 194-control composition, with complete source and tape certificates and a finite-trace equivalence to the pinned PA194QN construction. The equivalence is an ordinary mathematical proof formalizable in PA; no new Lean or Rocq kernel endpoint is claimed.

4.1 Literal identity and state budget

The tape alphabet is $\{0, 1\}$, the tape extends in both directions, and every cell is initially zero. The initial head coordinate is zero. The working controls are $0, \ldots, 193$, with initial control zero and separate terminal halt label 194. Every one of the 388 actions writes a bit and moves exactly one cell left or right. The dense table's SHA-256 is

```
7eb36aea69e5de46cc47fa7aab0191f191657ab500dafb808dc3c394640591b8
```

The native-label table starts at 74 and halts at 254. Its SHA-256 is

```
11613b0b12395fd3132bfe5038c47297685adf337459b138879bb8cdf790daba
```

The delivered bijection maps the complete native table to the dense table; all 194 working controls are graph-reachable.

The unfolded parent has 135 fixed controls and 67 AX readers. The seven Q-emitter folds and the added comparator fold are simultaneously compatible:

$$135 + 67 - 8 = 194.$$

The 32 local history/read cases are checked against the final simultaneous quotient. The arithmetic-box rows are unchanged. The separate literal symbolic checker derives its four list-sweep lemmas from the actual table, rather than calling the first checker's inherited arithmetic-box oracle.

4.2 Exact patch relative to QN

The frozen parent PA194QN has dense digest

`d1cf0f6e7a3c8f999d045694efc107dd1f9a03d45dd6c15e3d197ff7754c7231`.

The builder reconstructs its entire native and dense tables and source instruction map before applying the new transformation. The final native patch is exactly the following; every other row is identical to QN.

Control	Read	QN action	QNS action
28	0	$(0, R, 41)$	$(0, R, 330)$
38	1	$(1, R, 20)$	$(1, R, 30)$
41	0	$(0, R, 24)$	deleted
41	1	$(0, R, 25)$	deleted
70	0	$(0, R, 41)$	$(0, R, 330)$
310	1	$(1, R, 62)$	$(1, R, 330)$
330	0	absent	$(0, L, 31)$
330	1	absent	$(1, L, 46)$

The no-push action $(164, 1) \mapsto (1, R, 114)$ is already in QN and is retained. This avoids claiming it again as a new row change.

Before quotienting, added control 330 has actions $(0, R, 63)$ and $(1, L, 73)$. Incoming edges to old control 41 write zero and move right; incoming edges to raw 330 write one and move right. The immediately preceding cell therefore distinguishes the two histories. Folding $41 \to 330$ gives the displayed final actions and the four three-step local traces

$$330, 31, 21, 24; \quad 330, 46, 61, 25; \quad 330, 31, 47, 63; \quad 330, 46, 62, 73.$$

They cover histories zero/one and current bits zero/one, preserve the history cell and realize precisely the raw writes, moves and target controls. The other seven folds retain their full certificates in the presence of this additional fold.

4.3 An exact source equivalence

Both programs use logical registers $(N, T, W, A, D, H, X, Y, L)$, indexed from zero. Their physical allocation is $(6, 2, 0, 4, 3, 7, 8, 5, 1)$. QN has 3,654 primitive source instructions; QNS has 3,651. Their instructions numbered 0 through 3648 are literally equal, including every resolved target. Only the final five-instruction tail is replaced by two instructions:

Site	QN	QNS
3649	decrement X; zero $\to$ 3652, positive $\to$ 3650	increment H; $\to$ 3650
3650	increment H; $\to$ 3651	jump 0
3651	jump 3649	absent
3652	increment X; $\to$ 3653	absent
3653	jump 0	absent

The only edge from the common prefix into either tail is the zero branch of instruction 3645, which decrements H. Thus $H = 0$ on every tail entry arising from the designated initial source instruction 25. This is checked by a separate structural certificate that imports neither compiler nor physical executor.

The common instructions $0, \ldots, 10$ form a destructive equality test: on input $(H, X) = (u, v)$ they halt exactly when $u = v$; otherwise they terminate at instruction 11 with $H = X = 0$ and all other registers unchanged. Lockstep decrements detect simultaneous exhaustion; the remaining loops clear an unmatched counter. Induction on $u + v$ proves termination and the asserted result in PA.

At entry 3649 write $X = x$; the incoming invariant gives $H = 0$. QN's finite tail transfers X to H and then increments X, yielding comparator operands $(x, 1)$. QNS simply increments H, yielding $(1, x)$. They therefore both halt exactly when $x = 1$, and otherwise return to the identical complete store at instruction 11. The transfer loop terminates by induction on its initial value x.

More explicitly, the common Shift prefix at 3642 transfers incoming $T = t$ into X, then incoming $H = h$ into both T and X. Starting with arbitrary $(T, H, X) = (t, h, x)$, it reaches the tail with

$$(T, H, X) = (h, 0, x + t + h).$$

Both complete Shift blocks consequently halt exactly when $x + t + h = 1$; otherwise they leave $(T, H, X) = (h, 0, 0)$ and preserve every other register. At command boundaries $X = 0$, recovering precisely the original guard $t + h = 1$. No cached-head-only guard is being substituted.

For finite-trace transport, cut a source execution whenever it enters 3649 and follow that local block through the comparator to instruction 11 or halt. Replace each such terminating block by its counterpart. Outside these blocks the resolved instruction graphs are identical. The sole-entry check supplies $H = 0$ at every replacement, and every block has positive finite duration. Induction on the number of replacements gives primitive-recursive translations of finite halting traces in both directions, from any complete register store at source 25. PA proves the correctness of these translations. In particular,

$$\mathrm{PA} \vdash \mathrm{Halt}(\mathrm{source}_{\mathrm{QNS}}, 25) \leftrightarrow \mathrm{Halt}(\mathrm{source}_{\mathrm{QN}}, 25).$$

4.4 From the source to consistency

The complete blank computation of QNS reaches the clean source-25 frame after 58,004 physical steps, at native control 84 and head coordinate -2. The logical registers are $(2, 0, 0, 0, 0, 0, 0, 0, 0)$, followed by 26 one-valued spare registers. Both simulators compare the entire finite tape support and zero exterior, not a tape window. QN reaches that source frame after 58,154 steps.

The universal source-frame certificates cover 4,751 primitive outcomes, including both branches of every decrement, arbitrary natural primary registers, an arbitrary affine header gap, and the stated complete spare list interface. Together with positive progress and deterministic tape execution, they identify blank halting with halting of the corresponding source from its initial frame. The new source

equivalence therefore transports the supplied QN paper calibration:

$$\mathrm{PA} \vdash \mathrm{Halt}(\mathrm{PA194QNS}, \mathrm{blank}) \leftrightarrow \exists e\ \mathrm{Prf}_{\mathrm{PA}}(e, 0 = 1),$$

and hence

$$\mathrm{PA} \vdash \neg\,\mathrm{Halt}(\mathrm{PA194QNS}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA}).$$

The inherited QN calibration retains the separate diagonal-head calculus, the K/S exchange derivation for Q's curried-premise change, and the reserved-zero proof compiler for no-push MP. They are attributed in the reference fragments and are not proved again merely by executing the physical certificates. This transport needs no assumption that PA is consistent. Such an assumption is used only to conclude actual nonhalting from the biconditional.

4.5 Material difference and its limits

QNS changes the executing Shift protocol of QN, eliminating a counter transfer on every visit while retaining its arbitrary-store behavior. It changes the Q axiom emitter relative to the supplied sum-symmetric NS machine. These are recurring logical and runtime operations, not unreachable-row edits. Its complete action graph is absent from all 39 current 194-state graph classes at the pinned revision, even after state renaming, spanning-root change, reflection and bit interchange. The component techniques are inherited, and no sub-194 construction or global minimality result is claimed for this fallback.

5 Fresh verification and what it proves

Check	Fresh result
Working states / transitions	194 / 388
Raw controls / simultaneous folds	202 / 8
Local fold cases	32
Primitive source sites / outcomes	3,651 / 4,751
First symbolic executor	4,751 outcomes; 1,512,246 symbolic steps
Independent literal executor	4,751 outcomes; 1,106,186 symbolic steps
Generic list lemmas independently rederived	4
Minimum positive duration constant	3
Blank initialization	58,004 physical steps; entire tape equal
Prior 194-state classes excluded by exact graph comparison	39
Source prefix identical to QN	3,649 instructions
Exchange proof DAG	61 nodes; no hypotheses

The two symbolic checks have deliberately different trusted interfaces. The first executes the exact source contracts using inherited arithmetic block lemmas. The second imports no allocator or source compiler: it reconstructs the control-flow graph twice, proves four generic list sweeps from the literal rows, and executes every primitive outcome on symbolic complete tapes. Both quantify over arbitrary natural register values; they are not bounded tests on small register contents. The independent source-tail certificate checks the actual comparator rows, sole-entry invariant and identical source prefix without importing either physical executor.

The blank-start checks are finite, but they compare the whole finite tape support, head and control, with a zero exterior. The universal source simulation then applies from that exact seed. Every

simulated source step takes positive time. Determinism and this positive progress give the converse halting implication as well: a first physical halt cannot be passed by arbitrarily many positive simulated nonhalting steps.

These are executable mathematical certificates plus the stated ordinary proof. The JSON receipts explicitly record `kernel_checked=false`. No test is described as a completed Lean/Rocq proof of the PA or iPRA consistency equivalence, and no bounded nonhalting run is used to establish one.

6 Reduction research and a different numerical encoding

6.1 What was searched below 194

The first bounded comparison combines two AX grammars (primary and Q), four constructor coefficients (4, 5, 6, 8), four comparator/controller layouts and two final-return choices: 64 explicit combinations. For each, the program enumerates its one-bit incoming-history fold opportunities and checks compatible disjoint subsets against the final simultaneous table. The best physical count is 194. These are allocation screens, not blanket correctness certificates for every proposed command combination. Only the separately accepted constructions receive their complete interfaces and physical checks.

The SUMS and coefficient-5/6 combinations expose a concrete obstruction to adding previously published savings: both desired folds remove native control 41. Two new helpers cannot be paid for by deleting that one control twice. Their naive composition costs 195. A coefficient-eight construction also remains at 195 in the screened layout.

The new QNS table was additionally subjected to three explicit local contraction searches with radius six and a 256-step detour limit. The forward local abstraction contains 275,931 state/window contexts, starting from all 13-bit windows at native start 74. The searches find no accepted single same-head merge, no accepted two-square head-displacement merge, and no accepted adjacent-marker-bit merge. The latter two screens examine 32,367 and 32,173 eligible ordered pair/offset cases, with 1,565,312 and 914,435 action trials respectively. Complete output and the executable JavaScript are included.

These local abstractions overapproximate the encountered tape windows; the search tests a sufficient, very specific equivalence pattern. A negative result says nothing about more global tape recodings, changes to several helpers at once, a larger radius or a different proof calculus. It must not be described as a lower bound for arbitrary PA-consistency machines.

6.2 Exchange of other logical premises

The Q transformation exchanges the two premises of quantifier distribution. The same uniform exchange proof also permits

$$h_3^{\mathrm{e}}(D, A, H) = E(H, A) \to \big(M(H, D) \to M(A, D)\big),$$
$$h_4^{\mathrm{e}}(D, A, H) = M(A, H) \to \big(E(H, D) \to M(A, D)\big).$$

These change the membership-rewriting heads, rather than merely renaming their universally quantified parameters. Replacing any subset of h_3, h_4, h_6 by its exchanged form preserves numerical derivability by primitive-recursive translations in both directions. The independent free-algebra audit verifies all ten pushed AX entries and the exact guarded top for each proposed emitter. The 190-token words were then searched for different associations, repeated subtrees, compatible folds and parameter permutations. The final 330-second run made 1,727,732 proposal slots and accepted 342,713; its best

projected allocation was $134 + 71 - 8 = 197$. No candidate at or below 194 was found in that run. The proof of interderivability is useful independently of the allocation result.

6.3 TAG196N: a new formula-code involution

The secondary delivered construction changes the numerical syntax to

$$I'(A,B) = 4p(A,B), \quad M'(a,b) = 4p(a,b) + 1, \quad U'(a,B) = 4p(B,a) + 3, \quad E'(a,b) = 4p(a,b) + 2.$$

Only the quantifier and equality tags are exchanged; truth zero and contradiction one are fixed. This is a genuine change of the numbers constructed and compared during proof search. The stack and command scheduler retain their old pairing representation.

Let σ exchange 2 and 3 and fix 0 and 1. Define $F(0) = 0$, and for $n > 0$ write uniquely $n = 4p(a,b) + r$, $0 \le r < 4$, and put

$$F(n) = 4p(F(a), F(b)) + \sigma(r).$$

Both children are below n, so this is primitive-recursive course-of-values recursion. Strong induction gives the four constructor identities, with the *same* F applied to variable labels and formula codes. A second induction gives $F(F(n)) = n$. Thus F is a bijection of all natural codes; it fixes zero and one and preserves the distinct-label gate. In particular, the binder label in the recoded diagonal instance is $F(B)$, exactly the code of its recoded property. Recoding formulas alone would not prove that identity.

The translation is not order preserving: $F(2) = 3$ and $F(3) = 2$. Therefore one may not transport the side condition $d \le D$ by monotonicity. At a dominated-row leaf, use instead the supplied finite derivation of the *unrestricted* ρ instance. The component bounds for $I(A,B) = 4p(A,B)$ remain valid in the target. This supplies proof translations in both directions without a false numerical-order assumption. Translating a derivation and then compiling a fresh target command word is essential; an old word is not simply asserted to have the same execution.

The separate parameter screen completed 139,968 assignments. The final 330-second tag/Gamma search made 1,647,198 proposal slots and accepted 268,038; its best allocation remained 196. Proposal slots can repeat a grammar or make no change, so these counters are not counts of distinct machines. The accepted physical construction is verified independently of its search score.

The changed AX word has 202 tokens: its 15 quantifier tags each gain an increment, while its three equality tags each lose an increment. Exposing the existing triple-increment reader as a reusable AX macro is a useful allocation improvement. The delivered TAG196N budget is

$$135 \text{ fixed controls} + 71 \text{ AX readers} - 10 \text{ simultaneous folds} = 196.$$

It uses no-push MP and the original sum guard. Its source has 3,666 sites; all 4,767 universal primitive outcomes pass the first symbolic executor, with 1,535,770 symbolic operations and maximum 517. Complete blank initialization takes 58,320 steps. The arithmetic-box rows are unchanged dependencies of this check; the full second literal-checker run performed for QNS is not claimed for TAG196N. A separate no-compiler audit checks the entire 19-instruction no-push splice, and a free-constructor audit checks every emitted formula and the ten-entry stack interface. A further source-binding audit checks all 46 arithmetic-box rows against the frozen parent, all 61 constructor prefixes and four-increment bodies, and equality of the complete resolved primitive program before and after reassociation.

Its complete tables, deterministic builder, splice and emitter checks, mathematical recoding and command-compiler proofs, receipts and licenses are under `research/tag196n/`. The dense digest is

`e250b15fb2ad284cffaab5ed2842593f51d2ad7eec6c707aa42786c5ce3cfac6`.

The table is distinct from all 99 existing 196-state action graphs even under the stronger cosmetic comparison. This secondary result changes the arithmetic encoding, but **does not improve the 194-state bound**. The archive includes its universal mathematical proof separately from the bounded recoding calculations; those calculations are supporting audits, not proofs of involution for all natural numbers.

6.4 What transfers from the ZFC and ConciseIPRA work

The reusable ZFC ideas are exact implementation mechanisms: caller-history folding, simultaneous allocation checks, command-local field readers, compiler-supplied spare zeros and hereditary recoding of formulas *and* labels [6]. The no-push protocol here has that documented ancestry. The exact K215 external Lean/Rocq equivalence and its internal PA endpoint have different statuses; a proof for the former is not silently treated as a proof of the latter. Likewise, a ZFC command saving can conflict with the actual PA reader contexts and cannot simply be subtracted from the PA count.

ConciseIPRA shows how a small native calculus can derive familiar logic and how to remove redundant certificate machinery [7]. Its objective is the size of a complete Python proof checker, and its calibration target is iPRA. Both differ from a binary one-tape implementation searching PA refutations. Built-in substitution, binding and list operations still cost machine controls, and replacing the PA-strength proof supplier requires a new PA refutation translation. The useful lesson is to optimize the calculus, certificate interface and literal runtime together; no direct state saving or consistency-strength equivalence is inferred from a short Python checker.

7 Further research questions

1. Can a decoder grammar and its compatible contextual folds be optimized jointly, including fixed arithmetic readers? Minimizing only the raw grammar can worsen the final physical budget. The new code and axiom experiments provide additional exact search domains.
2. Can a larger local relation, or a simultaneous change of multiple helpers, remove a state that the radius-six, 256-step single-contraction model cannot remove? The present negative result only concerns that sufficient local model.
3. Can the unused zero-push work be traded for a different physical register/continuation interface that actually removes a control? Deleting source instructions alone has repeatedly left the state budget unchanged.
4. Can the formula-tag involution be allocated in 194 or fewer states, while retaining a checked command compiler? Its semantics are particularly simple, but the shared three-increment reader and history-fold constraints must be priced together.
5. Can the generic PA proof translations and literal simulation be internalized in the repository's intended iPRA endpoint? A useful next formal deliverable is the short QN–QNS tail equivalence, followed by the already identified common calibration obligations.

8 Reproduction and archive guide

Run `python3verify_all.py` from the extracted archive. This reconstructs the principal table in a fresh temporary directory, replays its source and literal-tape certificates, checks its identity and

graph novelty, and replays the finite exchange proof. Python 3.10 or later and its standard library are sufficient for the principal checks; assertions must remain enabled. Search replays additionally use Node.js or a C++ compiler as specified in `search/README.txt`.

The complete table is in `construction/machine/`; its source, instruction map, labels, local fold cases and fresh receipts are adjacent. `comparison/` contains all 54 packet records, all 39 frontier records, the full census and the 721 table byte presentations needed to audit it. `construction/references/` preserves the attributed mathematical dependencies, while `proofs/` contains the new exchange certificate and additional proof material. `search/` distinguishes accepted outputs from unaccepted candidates and bounded negative results.

Compile this report with `python3build_report.py`, or repeat `pdflatex` on `report/pa194_qns_research.tex` from its report directory until cross-references stabilize. The source files, both final PDFs, manifests and licenses are included. The supplementary `report/calculus_proofs.pdf` presents the full logical transports and the matching no-push command compiler. The archive is self-contained for the main construction and checks; only a fresh full repository census requires access to the pinned repository.

A Every existing 194-state graph

This is the pre-construction frontier at the pinned snapshot. Exact hashes, representative paths and every occurrence are in the accompanying CSV/JSON records. "Reported construction" describes the paper's evidence level; it does not assert completed iPRA formalization.

Name	Evidence status	Distinguishing construction
CS both-zero6	auxiliary	Known-zero X decrement at both source26 and final emitter exit; research composition.
CSZ-A3	auxiliary	CSZ plus comparator-return register3 and final-emitter register3 decrements.
CSZ-A5	auxiliary	CSZ plus comparator-return register3 and final-emitter register5 decrements.
CSZ-A6	auxiliary	CSZ plus comparator-return register3 and final-emitter register6 decrements.
Cached I-tree zero-jump seed	auxiliary	Alternate decoder plus cached controller and entry zero jump; search basis.
Gamma(D)-CSZA	auxiliary	Supplementary Gamma(D) emitter plus known-zero changes.
MPH	auxiliary	MP result cached in H while retaining original Shift; physical search input.
PA194CZ	auxiliary	Cached controller; entry known-zero decrement of register 6.
PA194altC	auxiliary	Cached controller on alternate 68-reader decoder.
Rejected CS node1-X	rejected	Native103/read1 becomes dec6; exact blank run halts at1361.
Rejected I-tree node1-X	rejected	Known-zero source rewrite fails blank bootstrap; exact halt at1361.
Rejected ZR node1-X	rejected	Register6 comparator-return variant; exact blank run halts at1361.
Rejected seven-zero scanner	rejected	Decoder local folds pass, but source-address scanner violates runtime contract.
M4R	reported construction	Reassociated coefficient4 constructor decoder; original resolved source graph.
M5	reported construction	Formula constructor coefficient5; explicit variable and proof recoding.
M6	reported construction	Formula constructor coefficient6; explicit variable and proof recoding.
PA194	reported construction	Separate diagonal head, 67-reader decoder with seven folds, original controller.
PA194-68/8	reported construction	Same separate-head source, reassociated 68-reader decoder with eight folds.
PA194-AD	reported construction	PA194 with physical A/D register exchange.
PA194A	reported construction	CS entry jump at source26 becomes decrement of known-zero register A.
PA194C	reported construction	MP result remains cached in H; Shift tests H=1; original ten-entry logical interface recompiled.
PA194CS	reported construction	Symmetric Shift guard; cached MP controller; contextual controller fold.
PA194CSZ = PA194X	reported construction	CS entry jump at source26 becomes decrement X; two headline names denote identical dense bytes.
PA194D	reported construction	Separate diagonal axiom uses Gamma(D), new proof selector, 69 readers and nine folds.
PA194F	reported construction	CS final emitter jump becomes decrement of known-zero register X.

Name	Evidence status	Distinguishing construction
PA194H (HS packet)	reported construction	No-push successful MP with cached-head-only Shift; asymmetric comparator.
PA194HA	reported construction	MP result cached in H with original sum guard, plus physical A/D placement exchange.
PA194HAZ	reported construction	HA plus known-zero final AX decrement; full startup and separate four-site source invariant.
PA194HS = PA194NS (cached)	reported construction	No-push successful MP with cached-head-only Shift; symmetric comparator. HS and lowercase NS packets give identical dense bytes.
PA194J	reported construction	Ten source jumps become known-zero decrements; three physical rows change.
PA194N	reported construction	Successful MP drops dummy-zero push, retains original Shift; revised command proof compiler.
PA194N5	reported construction	Coefficient-five formula constructors combined with no-push MP and original sum guard.
PA194N6	reported construction	Coefficient-six constructors combined with no-push MP; joint N5/N6 and standalone N6 packets give identical dense bytes.
PA194NS (sum)	reported construction	SUMS comparator with no-push MP. This is a different graph from the cached-only NS/HS machine despite the name collision.
PA194Q	reported construction	Quantifier-distribution curried premises exchanged; K/S proof translation.
PA194QN	reported construction	Exchanged curried quantifier-distribution premises combined with no-push MP and original sum guard.
PA194R	reported construction	Reassociated decoder of PA194F; 1111 source paths changed, same resolved source graph.
PA194SUMS	reported construction	Original sum guard with operands of destructive comparator reversed; one new history fold absorbs replacement trailer.
PA194ZR	reported construction	Eight comparator returns and final emitter return become known-zero register3 decrements.

B Complete principal transition table

For each state the two actions correspond to reading zero and one. An action $w\,D\,q'$ writes w, moves $D \in \{L, R\}$ and enters q'. State zero is initial and 194 is terminal. The two side-by-side state ranges are only a typesetting convention; together they give all 388 rows.

State	Read 0	Read 1	State	Read 0	Read 1
0	1 R 1	1 R 2	97	0 L 45	0 R 108
1	1 R 3	1 R 1	98	0 L 44	1 R 113
2	0 R 0	1 R 2	99	0 L 114	1 R 115
3	0 R 4	0 R 1	100	0 R 116	0 L 14
4	1 R 5	1 L 6	101	0 R 109	1 R 101
5	1 L 7	1 L 8	102	0 L 45	0 R 101
6	0 L 7	1 L 6	103	0 L 114	1 L 33
7	0 L 9	1 L 6	104	0 R 66	0 R 117
8	0 L 10	0 L 11	105	0 R 56	0 R 117
9	1 R 12	0 L 9	106	0 L 62	1 R 118
10	0 L 9	0 L 10	107	0 R 119	1 R 120
11	1 L 8	1 L 11	108	0 R 14	1 R 108
12	0 R 12	0 R 13	109	0 R 117	1 R 109
13	0 L 14	0 R 13	110	0 R 71	1 R 121
14	1 L 15	0 L 15	111	0 R 122	1 L 123
15	0 L 15	1 L 16	112	0 R 124	1 R 125
16	0 L 17	1 L 16	113	0 R 47	0 L 14
17	0 L 18	1 L 16	114	0 R 90	1 R 98
18	0 L 19	1 L 16	115	0 R 114	1 L 93
19	0 L 20	1 L 16	116	0 R 126	1 R 126
20	0 L 21	1 L 16	117	0 R 127	1 R 117
21	0 L 22	1 L 16	118	0 R 128	1 R 129
22	0 L 23	1 L 16	119	0 R 32	1 R 64
23	0 R 23	1 R 24	120	0 R 31	1 R 96
24	0 R 25	1 R 26	121	0 R 48	1 R 87
25	0 R 27	1 R 28	122	0 R 56	0 R 127
26	0 R 29	1 R 30	123	0 R 65	1 R 33
27	0 R 31	1 R 194	124	0 R 97	1 R 113
28	0 L 32	1 R 33	125	0 R 78	1 R 77
29	0 R 34	1 R 35	126	0 R 47	1 R 47
30	0 L 33	1 L 36	127	0 R 67	1 R 127
31	0 R 37	1 R 38	128	0 R 130	1 R 131
32	0 R 39	1 R 40	129	0 R 132	1 R 133
33	0 R 41	1 R 42	130	0 R 134	1 R 135
34	0 R 43	1 R 44	131	0 R 136	1 R 137
35	0 R 45	1 R 46	132	0 R 138	1 R 86
36	0 R 47	1 R 48	133	0 R 139	0 L 10
37	0 R 49	1 R 41	134	0 R 95	1 R 123
38	0 R 50	1 R 51	135	0 R 140	1 R 126
39	0 R 52	1 R 53	136	0 R 141	1 R 142
40	0 R 54	1 R 55	137	0 R 143	1 R 144
41	0 R 56	0 R 57	138	0 R 145	1 R 40
42	0 R 58	0 L 14	139	0 R 146	1 R 147
43	0 R 59	1 R 60	140	0 L 148	1 L 143
44	0 R 52	1 R 61	141	0 R 28	1 R 149
45	0 L 62	1 R 63	142	0 R 135	1 R 150
46	0 R 64	1 R 65	143	0 R 151	1 R 152

State	Read 0	Read 1	State	Read 0	Read 1
47	$0\,R\,66$	$0\,R\,67$	144	$0\,R\,153$	$1\,R\,154$
48	$0\,L\,64$	$1\,L\,68$	145	$0\,L\,130$	$1\,R\,47$
49	$0\,R\,69$	$1\,R\,51$	146	$0\,R\,31$	$1\,R\,50$
50	$0\,L\,70$	$0\,L\,15$	147	$0\,R\,155$	$1\,R\,146$
51	$0\,R\,71$	$0\,L\,10$	148	$0\,R\,156$	$1\,R\,157$
52	$0\,R\,72$	$1\,R\,73$	149	$0\,R\,158$	$1\,R\,159$
53	$0\,R\,74$	$1\,R\,75$	150	$0\,L\,159$	$1\,R\,160$
54	$0\,R\,76$	$1\,R\,77$	151	$0\,R\,40$	$1\,L\,161$
55	$0\,R\,78$	$1\,R\,79$	152	$0\,R\,162$	$1\,R\,163$
56	$0\,L\,80$	$0\,R\,1$	153	$0\,R\,164$	$1\,R\,164$
57	$0\,R\,81$	$1\,R\,57$	154	$0\,R\,165$	$1\,R\,166$
58	$0\,L\,72$	$1\,L\,65$	155	$0\,R\,121$	$1\,R\,111$
59	$0\,R\,82$	$0\,L\,7$	156	$0\,R\,75$	$1\,R\,165$
60	$0\,R\,83$	$1\,R\,84$	157	$0\,R\,167$	$1\,R\,165$
61	$0\,R\,41$	$1\,R\,85$	158	$0\,R\,168$	$1\,R\,169$
62	$0\,R\,86$	$1\,R\,87$	159	$0\,R\,170$	$1\,R\,157$
63	$0\,R\,88$	$0\,L\,71$	160	$0\,R\,135$	$1\,R\,130$
64	$0\,R\,30$	$1\,R\,87$	161	$0\,R\,138$	$1\,R\,135$
65	$0\,R\,69$	$1\,R\,36$	162	$0\,R\,171$	$1\,R\,172$
66	$0\,L\,80$	$0\,R\,2$	163	$0\,R\,173$	$1\,R\,172$
67	$0\,R\,89$	$1\,R\,67$	164	$0\,L\,174$	$1\,R\,175$
68	$0\,R\,90$	$1\,R\,91$	165	$0\,L\,162$	$1\,L\,176$
69	$0\,R\,56$	$0\,R\,81$	166	$0\,R\,161$	$1\,R\,176$
70	$0\,R\,65$	$1\,R\,92$	167	$0\,R\,95$	$1\,R\,177$
71	$0\,R\,41$	$0\,L\,13$	168	$0\,R\,134$	$1\,R\,178$
72	$0\,R\,82$	$1\,R\,68$	169	$0\,R\,150$	$1\,R\,163$
73	$0\,R\,48$	$1\,R\,93$	170	$0\,R\,179$	$1\,R\,180$
74	$0\,R\,41$	$1\,R\,94$	171	$0\,R\,167$	$1\,R\,140$
75	$0\,R\,95$	$1\,R\,96$	172	$0\,R\,134$	$1\,R\,145$
76	$0\,R\,97$	$1\,L\,98$	173	$0\,R\,181$	$1\,R\,40$
77	$0\,R\,48$	$1\,R\,76$	174	$0\,R\,175$	$1\,R\,182$
78	$0\,R\,92$	$1\,R\,99$	175	$0\,R\,167$	$1\,R\,145$
79	$0\,R\,48$	$1\,R\,100$	176	$0\,R\,183$	$1\,R\,173$
80	$1\,R\,80$	$1\,L\,15$	177	$0\,R\,86$	$1\,R\,61$
81	$0\,R\,101$	$1\,R\,81$	178	$0\,R\,184$	$1\,R\,165$
82	$0\,L\,45$	$0\,R\,89$	179	$0\,R\,134$	$1\,R\,181$
83	$0\,R\,102$	$1\,R\,93$	180	$0\,R\,185$	$1\,R\,185$
84	$0\,R\,102$	$1\,R\,103$	181	$0\,R\,167$	$1\,R\,151$
85	$0\,R\,104$	$0\,L\,14$	182	$0\,R\,151$	$1\,R\,130$
86	$0\,R\,105$	$1\,L\,70$	183	$0\,R\,164$	$1\,R\,186$
87	$0\,R\,103$	$0\,L\,14$	184	$0\,R\,140$	$1\,R\,187$
88	$0\,R\,106$	$1\,R\,107$	185	$0\,R\,188$	$1\,L\,189$
89	$0\,R\,108$	$1\,R\,89$	186	$0\,R\,190$	$1\,R\,40$
90	$0\,R\,66$	$0\,R\,109$	187	$0\,R\,32$	$1\,R\,189$
91	$0\,R\,110$	$0\,L\,87$	188	$0\,R\,47$	$1\,R\,191$
92	$0\,R\,30$	$1\,R\,58$	189	$0\,R\,148$	$1\,R\,192$
93	$0\,R\,82$	$1\,L\,111$	190	$0\,R\,40$	$1\,R\,138$
94	$0\,R\,99$	$0\,L\,14$	191	$0\,R\,174$	$1\,R\,47$
95	$0\,R\,64$	$1\,R\,112$	192	$0\,R\,182$	$1\,R\,193$
96	$0\,R\,111$	$1\,R\,74$	193	$0\,R\,161$	$1\,R\,138$

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